

SHOW LOW, AZ, USA

WMO: 723747

Lat: 34.264N

Lon: 110.008W

Elev: 6411

StdP: 11.60

Time Zone: -7.00 (W07)

Period: 05-19

WBAN: 03101

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.5	13.9	-5.9	5.1	33.9	-1.8	6.3	34.7	34.2	42.9	29.0	41.9	2.9	80	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.7	90.0	56.2	87.7	56.2	84.2	55.8	62.6	74.6	61.7	74.3	60.8	73.5	10.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.5	96.6	65.5	57.4	89.4	65.1	56.7	87.1	65.1	31.8	75.9	31.0	73.8	30.2	73.8	68.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
31.5	26.5	23.1		DB	0.4	94.0	5.6	1.7	-3.7	95.2	-7.0	96.2	-10.1	97.1	-14.2	98.3
				WB	-0.5	64.8	5.3	1.7	-4.4	66.0	-7.4	67.0	-10.4	68.0	-14.2	69.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	33.1	37.8	44.5	50.2	57.4	69.2	71.7	69.3	64.0	53.5	43.6	33.8	(6)
	DBStd	15.04	7.99	8.17	6.93	6.78	7.03	5.00	3.76	3.82	4.87	6.84	7.80	8.58	(7)
	HDD50	1924	525	345	191	79	21	0	0	0	1	43	215	504	(8)
	HDD65	5146	990	763	634	445	248	19	1	8	71	357	641	969	(9)
	CDD50	2800	0	2	21	84	249	576	671	598	422	152	24	1	(10)
	CDD65	548	0	0	0	0	11	145	208	141	42	1	0	0	(11)
	CDH74	6258	0	0	0	17	325	2208	1853	1263	552	40	0	0	(12)
	CDH80	1951	0	0	0	1	48	817	665	344	75	1	0	0	(13)
	WSAvg	9.0	7.4	9.6	10.3	11.6	10.8	10.2	8.0	7.1	8.1	8.2	8.4	8.4	(14)
	PrecAvg	13.9	1.1	0.9	0.8	0.6	0.5	0.4	2.2	2.8	1.5	1.2	0.9	1.3	(15)
Precipitation	PrecMax	21.8	4.2	2.6	2.2	1.9	4.1	1.1	4.6	5.4	4.0	3.2	2.6	3.7	(16)
	PrecMin	7.8	0.1	0.0	0.0	0.0	0.0	0.0	0.4	0.9	0.1	0.0	0.0	0.0	(17)
	PrecStd	2.8	1.1	0.7	0.7	0.5	0.9	0.3	1.3	1.3	1.0	0.8	0.7	1.1	(18)
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.6	64.3	71.6	75.5	84.1	91.5	92.7	89.7	84.4	78.7	71.8	61.3	(19)
		MCWB	40.0	42.9	45.5	48.0	52.7	55.3	56.5	58.3	57.0	53.3	47.5	40.5	(20)
	2%	DB	54.4	62.6	67.5	72.7	81.2	90.2	89.9	86.2	81.9	74.6	66.3	56.7	(21)
		MCWB	38.9	41.6	44.5	46.5	50.8	54.9	57.7	58.0	55.8	50.9	45.1	39.8	(22)
	5%	DB	50.3	57.1	63.9	70.1	78.5	88.0	87.5	83.8	80.7	72.1	63.3	52.3	(23)
		MCWB	37.1	39.8	42.5	45.5	49.3	54.3	57.6	58.3	55.4	49.6	43.8	38.7	(24)
	10%	DB	46.1	53.7	60.9	66.2	73.5	84.3	83.8	81.5	77.3	69.5	59.0	47.7	(25)
		MCWB	35.8	38.3	40.8	43.9	47.9	53.2	57.5	58.3	54.5	49.0	41.7	36.9	(26)
	0.4%	WB	44.4	45.5	46.7	50.3	54.9	59.9	64.8	64.2	61.4	58.3	49.0	44.9	(27)
		MCDB	49.7	60.3	64.3	70.5	79.6	76.2	74.0	75.1	74.0	66.8	64.6	52.1	(28)
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.2	42.4	45.1	48.1	52.5	57.8	63.1	62.5	59.8	54.8	46.9	42.1	(29)
		MCDB	50.4	56.8	64.1	69.6	74.9	79.5	74.5	75.0	72.1	67.4	61.0	51.9	(30)
	5%	WB	38.8	40.8	43.5	46.2	50.9	56.5	62.0	61.8	58.8	52.8	45.1	40.1	(31)
		MCDB	47.5	54.9	61.1	66.8	72.7	80.0	74.0	74.8	71.7	65.0	60.1	50.0	(32)
	10%	WB	36.8	39.0	42.0	44.5	49.3	55.1	61.1	61.0	57.4	50.5	43.3	38.0	(33)
		MCDB	45.3	51.0	58.3	64.1	71.3	79.7	73.5	73.9	71.5	64.7	57.3	46.3	(34)
Mean Daily Temperature Range	5% DB	MDBR	21.9	22.6	25.6	27.5	27.7	29.3	23.7	23.1	24.9	25.9	24.8	20.5	(35)
		MCDBR	27.2	29.3	30.6	31.9	31.6	32.0	29.1	27.0	28.0	29.7	28.9	26.0	(36)
		MCWBR	15.8	15.4	14.7	15.0	13.7	12.1	9.0	8.6	10.5	13.6	14.7	14.7	(37)
	5% WB	MCDBR	22.3	25.9	27.8	29.0	28.2	27.8	21.3	21.2	21.9	23.4	25.3	21.6	(38)
		MCWBR	14.0	14.0	13.6	13.8	12.1	10.4	7.3	7.6	8.5	12.0	13.5	13.1	(39)
Clear-Sky Solar Irradiance	taub		0.236	0.240	0.263	0.271	0.282	0.296	0.344	0.344	0.300	0.269	0.244	0.236	(40)
	taud		2.667	2.682	2.583	2.553	2.567	2.520	2.488	2.445	2.567	2.626	2.656	2.671	(41)
	Ebn at Noon		319	328	324	323	318	311	296	295	306	311	313	314	(42)
	Edh at Noon		22	24	30	32	32	33	34	35	30	25	22	21	(43)
All-Sky Solar Radiation	RadAvg		1046	1348	1821	2261	2514	2599	2128	1989	1847	1525	1174	933	(44)
	RadStd		91	116	108	104	152	137	125	86	76	121	67	61	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (10 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	N/A	-173	+128	+50

Nomenclature: See separate page